MATHEMATICAL RESEARCH NOTE REPAIRED VERSION 0.2 | 26 OCTOBER 2026

Two Armies of One

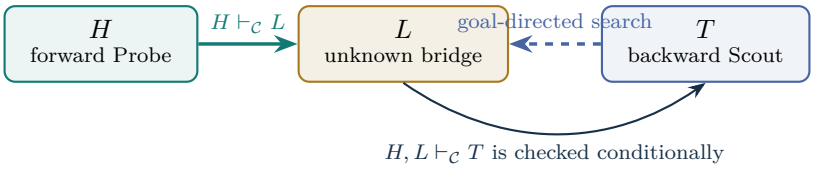
Certified Scout–Probe Rendezvous in Proof Search

Corrected guarantees, witness-relative middle cuts,
and the first-settlement barrier

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Mathematical exposition and ATLAS-style repair: **ChatGPT (OpenAI)**

PRELIMINARY — REPAIRED, NOT FORMALLY CERTIFIED

Abstract. Exactly two search agents are considered: a backward Scout seeking a checked conditional continuation to a target T, and a forward Probe deriving verified consequences of a frozen hypothesis bank H. An accepted Probe proof $H \vdash_{\mathcal{C}} L$ can close a compatible Scout proof $H, L \vdash_{\mathcal{C}} T$; that composition is sound. For a *fixed* Scout certificate, unbounded fair Probe enumeration finds a rendezvous at L exactly when $H \vdash_{\mathcal{C}} L$. For a *route-changing* Scout, fair two-sided enumeration finds a rendezvous whenever a finite, checker-compatible *admissible triple of certificates* exists. The nontriviality test is made decidable on the supplied finite certificates, not asserted to be decidable for a bare formula. A target-free proper proof cut (without any claim of semantic necessity) is distinguished from a chronological meeting before direct settlement, which is *not* guaranteed. The readiness frontier uses actual open proof leaves. No speedup, finite time bound, working agent implementation, formally machine-checked theorem, or independent certification is claimed.



The dashed arrow is search direction, not a reverse logical inference.

1 Question, architecture, and claim levels

There are *two armies of one*: one Scout and one Probe. Their meeting point need not be specified in advance. Both may search indefinitely, and matching and checking are operations of their common protocol, not additional search agents. The word *certificate* has a strict meaning: a finite proof object accepted by the stated verifier. An unverified candidate is only a proposal.

Three claims must remain distinct:

1. **Composition soundness:** two compatible certificates certify T.
2. **Conditional eventual discovery:** a finite eligible matching pair is eventually detected by a search that is fair, persistent, and allowed to continue.
3. **A useful early meeting:** an eligible lemma is found before either agent independently proves T, perhaps at less total cost than a matched comparator.

Only the first two are established under explicit assumptions. The third requires a separate result or experiment.

2 Formal setting and certificate discipline

Fix a countable effectively encoded propositional or first-order presentation $\mathcal{C}$ whose *only proper inference rule is modus ponens.* Assume its axiom instances are decidably recognizable, the authorized hypothesis bank H is decidable, and a sound terminating verifier V recognizes valid finite $\mathcal{C}$-certificates. Checking must include axiom and hypothesis membership, formula equality, ordered MP ports, permitted variable conventions, and imports. Such finite strings and proof objects can be fairly enumerated; an enumerator is not assumed to decide theoremhood.

A proof certificate is a finite acyclic directed graph with a specified conclusion occurrence, labeled leaves (axiom, H-hypothesis, or designated open assumption), ordered MP-premise ports, and any necessary imported subcertificates or valid receipts. Before an ancestry test, *trim* all nodes not feeding the distinguished conclusion. For the structural tests below, imports are either fully expanded or accompanied by a verified, inspectable provenance closure. A ledger cannot invent missing proofs, rewrite H, or silently substitute a different calculus or verifier.

Definition 2.1 (Probe and Scout certificates)**.** A Probe certificate π_L is a finite checked proof $H \vdash_{\mathcal{C}} L$ with no unresolved assumptions. A Scout certificate ρ_L is a finite checked *conditional* proof $H \cup \{L\} \vdash_{\mathcal{C}} T$ whose trimmed conclusion ancestry contains a distinguished *open leaf* labeled L. The Scout has not thereby proved $H \vdash_{\mathcal{C}} T$. These are proof occurrences, not merely matching formula names.

Definition 2.2 (Certified rendezvous)**.** A rendezvous at L occurs when a checked π_L and checked ρ_L are available in compatible contexts, their import and occurrence receipts pass a terminating compatibility check, replacement of the open L occurrences by π_L is legal, and the completed $H \vdash_{\mathcal{C}} T$ proof passes V. A planned reverse edge is *not* a proof of a premise from its conclusion.

Remark 2.3 (The original calculus remains a dependency)**.** An MP-only presentation $F^\star$ may be instantiated as $\mathcal{C}$. The claim that a proof in $F^\star$ can be decoded into an original first-order presentation F belongs to a *separate* translation theorem and audit. No result in this note independently proves that translation correct.

3 Soundness of certificate composition

Theorem 3.1 (Certified composition)**.** *Let π_L be a checked $H \vdash_{\mathcal{C}} L$ certificate and ρ_L a checked $H, L \vdash_{\mathcal{C}} T$ conditional certificate. If their contexts, imports, and occurrences are compatible and finite splicing is permitted by the proof format, an algorithm constructs a finite checked $H \vdash_{\mathcal{C}} T$ proof.*

Proof. Trim ρ_L to the ancestry of its concluding T. Replace each distinguished open leaf labeled L by a fresh copy of π_L or by a checker-permitted reference to its complete receipt. Since the original edges point from premises to conclusions and the inserted proof is closed over H, the result is a finite acyclic derivation with leaves only from H or the original axiom stock. Each MP application keeps its original ordered inputs. Run V on the assembled certificate. Its acceptance certifies $H \vdash_{\mathcal{C}} T$. □

Corollary 3.2 (Conditional return translation)**.** *If there additionally exists a separately validated total decoder D mapping checked $\mathcal{C}$-proofs of $H \vdash_{\mathcal{C}} T$ into checked proofs of $H \vdash_F T$, then applying D to the composite proof yields such an F-proof, provided the final F checker accepts it.*

Remark 3.3 (Why assumption scope is not cosmetic)**.** In a calculus whose rules include assumption-sensitive generalization, blind proof insertion can invalidate a rule because its premise acquires new open support. The theorem intentionally concerns the specified MP-only certificate format; it does not license unsupported proof splicing in an arbitrary calculus.

4 One fixed Scout route: the exact eventuality theorem

Definition 4.1 (Continuing fair Probe)**.** A Probe is *fairly exhaustive* if it eventually generates or otherwise records a checked certificate of each H-provable formula (for example, by dovetailing all finite candidate proof encodings). Accepted records persist, every relevant pair is eventually examined by the matching routine, and the protocol continues without a finite time cutoff or early stop.

Theorem 4.2 (Fixed-Scout rendezvous equivalence)**.** *Fix one accepted Scout certificate $\rho_L : H, L \vdash_{\mathcal{C}} T$ with an actual used open L leaf. Under a continuing fair Probe, persistent certificates, compatible receipt policy, and fair matching, the protocol eventually certifies a rendezvous* at *L if and only if $H \vdash_{\mathcal{C}} L$.*

Proof. If $H \vdash_{\mathcal{C}} L$, a finite valid H-proof of L exists. The fair Probe records an accepted one after finitely many of its own search stages. The matcher eventually compares that proof with ρ_L; their assumed compatibility and Theorem 3.1 give the certified meeting. Conversely, a certified rendezvous includes a checked H-proof of L, so $H \vdash_{\mathcal{C}} L$. Neither direction provides a bound on the meeting stage. □

Example 4.3 (A stranded Scout despite a provable target)**.** In the classical propositional fragment take

$$H = \{p, p \to t, q \to t\}, \qquad T = t, \qquad L = q.$$

The Probe can establish t using $p, p \to t$. The Scout's conditional route from q to t is checked using $q \to t$, but $H \nvdash q$: the valuation $p = t =$ true and $q =$ false satisfies H and refutes q. Thus that fixed Scout cannot meet the Probe at q. The provability of T alone is insufficient for an arbitrary fixed L.

5 A Scout that changes routes: conditional completeness

Definition 5.1 (Fair two-sided proof search)**.** The Scout fairly enumerates all finite, accepted conditional objects (L, ρ) using an open L leaf to derive T from H, L. The Probe fairly enumerates all finite checked proofs π from H, or at least one accepted proof for every derivable conclusion when no certificate-sensitive filter is used. For a predicate depending on the *particular Probe proof*, full finite-certificate coverage is required. Certificates persist, and a dovetailed matcher fairly tests jointly available triples (L, π, ρ). Neither search is allowed to terminate upon an unrelated proof if an eventual detection result for a specified witness is being asserted.

Definition 5.2 (Computable witness admissibility)**.** An admissibility rule $\mathsf{Adm}(L, \pi, \rho)$ is a total terminating algorithm on an explicitly supplied finite triple. It may test formula syntax, contexts, scope closure, dependency graphs, target occurrences, and provenance receipts. It is *not* a decision procedure for the unbounded claim "there exists some admissible proof triple for L."

Theorem 5.3 (Witness-filtered eventual rendezvous)**.** *Assume the setting above with fair two-sided* full-certificate *enumeration, persistent records, fair matching, and continued unbounded execution. If there exists a compatible finite triple* (L, π_L, ρ_L) *such that*

$$\pi_L : H \vdash_{\mathcal{C}} L, \qquad \rho_L : H, L \vdash_{\mathcal{C}} T, \qquad \mathsf{Adm}(L, \pi_L, \rho_L) = \text{true},$$

then the two-agent protocol eventually detects and certifies an Adm*-admissible rendezvous (possibly via a different qualifying triple).*

Proof. The witness consists of finitely encoded proof objects. Fair enumeration eventually places its Scout and Probe records in persistent storage. Fair triple matching eventually tests that pair. Because Adm is terminating and accepts the given compatible triple, and splicing is checked, Theorem 3.1 establishes the rendezvous. The argument guarantees no order relative to a separate direct proof of T. □

5.1 The deliberately weak guarantee at $L = T$

If $H \vdash_{\mathcal{C}} T$ and the policy permits $L = T$, the Scout has a one-leaf conditional certificate $H, T \vdash_{\mathcal{C}} T$ while the Probe eventually discovers a checked $H \vdash_{\mathcal{C}} T$ proof. This is a valid but *degenerate* meeting: Probe completion of T already settles the target. It is incorrect to promote this statement into a guarantee of a useful internal bridge. Conversely, when $L = T$ is excluded, target provability by itself supplies *no* witness to Theorem 5.3.

6 What can "somewhere in the middle" mean?

Definition 6.1 (Target-free structural witness cut)**.** For fully expanded, compatible finite certificates define $\mathsf{Mid}(L, \pi, \rho)$ to hold when all the following finite checks pass:

(i) $L \neq T$ and $L \notin H$ as literal formula membership;

(ii) the trimmed Probe certificate π ends at L, contains at least one proper inference step, and contains *no occurrence of* T *anywhere in its disclosed nodes*;

(iii) the trimmed Scout certificate ρ contains a designated open leaf L in the concluding T's ancestry, with a nonempty chain containing at least one proper inference step from that leaf to the final T;

(iv) exact contexts, supports, ordered MP ports, imported receipts, and the complete splice pass verification.

This is a *certificate-relative target-free cut*—not a declaration that the meeting is efficient or temporally earlier than direct settlement.

The predicate Mid is decidable *given the finite expanded witnesses.* A program tests syntactic equality, ancestry in finite acyclic graphs, inference-node counts, and proof checking. This does not assert that $\{L : \exists \pi, \rho\ \mathsf{Mid}(L, \pi, \rho)\}$ is decidable.

Proposition 6.2 (Conditional detection of a target-free cut)**.** *If one compatible triple satisfies* Mid*, then the continuing, fully fair two-agent protocol using* $\mathsf{Adm} = \mathsf{Mid}$ *eventually detects such a certificate-relative cut and completes a checked proof of* T*.*

Proof. The four clauses of Definition 6.1 give a total predicate on the supplied finite triple. Apply Theorem 5.3. □

Proposition 6.3 (A certified DAG cut supplies a bridge)**.** *Suppose a finite, verified, trimmed proof DAG of* $H \vdash_{\mathcal{C}} T$ *contains an internal occurrence* v *labeled* $L \notin H \cup \{T\}$*. Suppose* v *has at least one inference ancestor, lies on a directed path to the final* T *through a proper inference step, and its ancestor sub-DAG has no node labeled* T*. Then severing* v *yields compatible Probe and Scout certificates satisfying* Mid*.*

Proof. The ancestry at v is itself an H-proof π_L; expand imports and keep all premise dependencies. In a copy of the original DAG, replace v by a distinguished open L leaf and discard any now-unneeded nodes; this is a checked conditional proof ρ_L of T from H, L. The designated leaf remains on a dependency path to T. All four finite checks in Definition 6.1 follow from the hypotheses; splicing restores a valid proof of T. □

Remark 6.4. This proposition does not assert that every provable target has such a cut. It gives a sufficient structural criterion for the existence of a witness. It also does not impose a cost advantage on the agents that find it.

6.1 Two precise counterexamples to overclaiming

Example 6.5 (A false midpoint: a target-first detour)**.** Work in a classical propositional MP-only Hilbert presentation with ordinary tautology axioms, and set

$$H = \{p, p \to t\}, \quad T = t, \quad U = (q \to q), \quad L = t \wedge U.$$

There is a checked Probe derivation of t from $p, p \to t$, followed by a derivation of U and a derived conjunction-introduction proof of L. A Scout can use the conditional leaf L and the propositional theorem $(t \wedge U) \to t$ to derive t. Thus $L \notin H$, $L \neq T$, and the spliced proof has nonzero work on both sides. *Nevertheless* T *occurred in the Probe proof before* L*.* The old syntactic and proper-cut criteria wrongly classified this as an informative middle meeting. Condition (ii) of Mid rejects the triple.

Example 6.6 (A target-free certificate cut need not be early)**.** Let $H = \{p, p \to \ell, \ell \to t\}$ and $T = t$, $L = \ell$. The one-step MP proof of ℓ and conditional MP continuation from ℓ to t satisfy Mid. But a fair search schedule can complete and record an entirely separate proof of t *before* its Probe and Scout enumerations jointly expose that cut. Fairness constrains eventual inclusion, not the relative order of discoveries. So Mid is *not* a guarantee of an early runtime rendezvous.

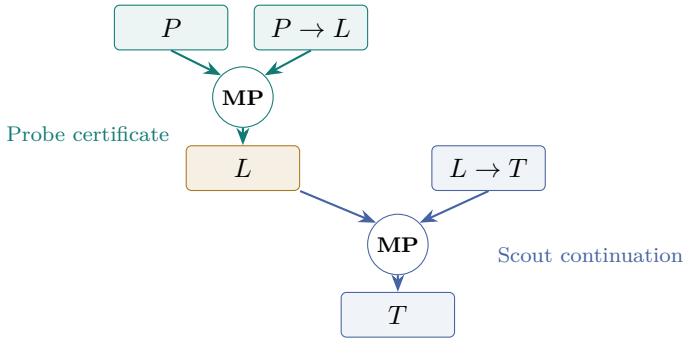
Example 6.7 (A target-free cut can be logically inessential)**.** Let $H = \{t\}$ and $T = t$. Choose $L = (q \to q)$, which has an MP-based closed proof using only propositional axiom instances involving q. Take such a proof with at least one MP step as the Probe certificate; it contains no occurrence of t. For the Scout, use the axiom $t \to (L \to t)$, the hypothesis t, and MP to obtain $L \to t$; then use the open L and MP to obtain t. Both selected proof branches contain proper inference steps, and the triple passes Mid. Nevertheless $H \vdash_{\mathcal{C}} T$ has the one-line hypothesis proof, without any rendezvous. Thus even the repaired structural predicate establishes only a *target-free certificate cut*, not that L is mathematically necessary or operationally worthwhile.

Remark 6.8 (Why global logical necessity is the wrong requirement)**.** For any successful rendezvous the composition theorem already gives $H \vdash_{\mathcal{C}} T$. It would therefore be self-defeating to define a useful bridge by requiring $H \nvdash_{\mathcal{C}} T$ when L is omitted: that condition cannot coexist with a successful proof of T from H. Use a specified proof DAG, chronology, or a charged comparison to describe usefulness, never an assertion that L is globally indispensable to consequence.

Chronological first-settlement distinction. Let s_L be the actual step when compatible L-certificates are first joined, and s_T the step at which an unassisted branch of the declared run first produces an accepted $H \vdash_{\mathcal{C}} T$ proof. A *pre-settlement* meeting would require $s_L < s_T$, with the event history and cost model fixed in advance. Theorem 5.3 makes no such assertion. Some runs terminate at s_T before any eligible rendezvous is detected; those runs do not satisfy the continuing-search premise used for the eventual-admissible-witness claim.

7 A correctly drawn two-army example

Let $H = \{P, P \to L, L \to T\}$. The following is a two-MP proof with a precise meeting occurrence and two proper inference nodes:



The Probe certifies $H \vdash_{\mathcal{C}} L$. The Scout presents a checked conditional proof $H, L \vdash_{\mathcal{C}} T$ using the L leaf and $L \to T$. At rendezvous the certified L occurrence closes the Scout leaf. The diagram does not pretend that L follows from T.

8 One Scout with several genuinely open obligations

A conditional proof of T may require a *finite set* of open leaves, not just one. No third agent is required: the same Probe can certify several lemmas over time.

Definition 8.1 (Exact obligation support)**.** For a checked, trimmed conditional certificate ρ whose declared open labels lie in a finite set O, define

$$\mathsf{Used}(\rho) = \{L : \text{an open leaf labeled } L \text{ occurs in the ancestry of its designated final } T\}.$$

This is the actual *syntactic* obligation support of that proof occurrence, not the set of all formulas that were offered to it. The support may not be semantically minimal across all alternative proofs.

Proposition 8.2 (Exact finite discharge)**.** *If $\rho : H \cup O \vdash_{\mathcal{C}} T$ is a finite checked conditional proof and every $L \in \mathsf{Used}(\rho)$ has a compatible checked H-proof, then finite proof splicing yields a checked H-proof of T. Under a continuing fair Probe, persistence, and fair matching, any specified finite* fully dischargeable *set $\mathsf{Used}(\rho)$ is eventually completed.*

Proof. Only the open leaves in the trimmed final ancestry matter. Replace their occurrences by the compatible H-proofs, check the finite composite, and obtain $H \vdash_{\mathcal{C}} T$. For eventuality, each of finitely many proof certificates is eventually recorded; after their latest arrival, a fair matcher eventually performs the composition. □

For a *fixed finite family* of already checked conditional routes $\rho_1, \dots, \rho_m$ over one common verifier/context with compatible receipts, put $O_i = \mathsf{Used}(\rho_i)$ and let B denote the set of currently available certified H-conclusions. Within this particular family, and under the compatibility assumption,

$$\mathsf{Ready}(B) = \bigvee_{i=1}^{m} [O_i \subseteq B].$$

Deleting a set O_j that strictly contains another listed O_i does not change this readiness predicate. Thus the distinct inclusion-minimal O_i form an antichain *for the selected finite family.* If receipt compatibility varies between routes, the correct frontier uses context-qualified proof receipts rather than plain formula-set inclusion. This is not the same object as a history-unfolded SIC frontier.

9 A fair protocol and its stopping policy

Schematic two-agent protocol (no experimental execution claimed).

1. Freeze $\mathcal{C}, H, T, V$, the admissibility predicate, receipts, schedule, and charged costs.
2. Dovetail Scout enumeration of finite checked conditional certificates (single leaves or declared finite sets). Keep rejected proposals in CAND, not in a certified lemma bank.
3. Dovetail Probe enumeration of finite checked H-proofs. Append checker-accepted certificates to a persistent provisional ledger; do not silently mutate the frozen hypothesis bank.
4. Fairly enumerate jointly available certificate pairs/triples; check exact contexts and imports, then apply Adm on the *full witness.*
5. On successful splicing and a final V pass, record a certified $H \vdash_{\mathcal{C}} T$ conclusion. Invoke a return decoder only if that decoder has separately passed its audit.
6. If the goal is guaranteed detection of an eligible witness, continue searching until it is detected; an earlier, unrelated H-proof of T must not stop that special search. In a normal first-settlement run, report the first verified result and do *not* claim the same eventual eligible-witness guarantee.
7. Under a finite budget, if no accepted terminal result is produced, report UNKNOWN rather

than treating failure to meet as a refutation.

A fair rescue enumeration may coexist with heuristics, but it must not be starved if its eventuality guarantee is invoked. Each proof check, proposal, import, match, storage operation, and final verification must be charged under an explicitly declared machine and budget model. For example,

$$W = W_{\text{Scout}} + W_{\text{Probe}} + W_{\text{check}} + W_{\text{match}} + W_{\text{store}} + W_{\text{compose}} + W_{\text{decode}} + W_{\text{final}}.$$

The above mathematical assumptions do not provide a general quantitative upper bound for W, nor do they establish a speedup over an equally resourced prover allowed both forward and backward search.

10 Status of every research claim

Claim	Correct status
Sound splicing of compatible MP certificates	Proven conditionally (Theorem 3.1).
Fixed-route meeting iff $H \vdash_{\mathcal{C}} L$	Proven under continuing fair Probe and matching (Theorem 4.2).
Changing-route Scout finds an existing eligible triple	Proven with full-certificate fairness, persistence, and continuing execution (Theorem 5.3).
Guaranteed useful midpoint from $H \vdash_{\mathcal{C}} T$ alone	**Not proved; false as a claimed consequence of the assumptions.** Only $L = T$ is universally available if allowed.
Certificate-relative target-free proper cut	Decidable for a given finite triple; existence requires an additional witness (Definition 6.1, Proposition 6.2).
Meeting before first direct proof of T	**Not guaranteed**; depends on actual schedule and run (Example 6.6).
Semantic necessity of the bridge	**Not guaranteed**, even for a target-free cut (Example 6.7).
Strictly lower fully charged cost	Open; no measurements presented.
Decoding from $F^\star$ to original calculus F	Dependent on a separate verified decoder and final original checker.
Working two-agent prover / formal proof assistant certification	Not supplied or claimed.

11 Correction history and open problems

Version 0.2 responds to both 26 October 2026 ATLAS-style paper audits. It replaces the misleading formula-only "middle" test, rejects a stronger but still spurious target-first proper-cut example, and separates certificate structure from chronological runtime advantage. The precise correction ledger accompanies this manuscript as a separate file.

The open mathematical questions are whether natural infinite target families admit useful internal cuts with controllable costs, whether an independently scheduled Scout and Probe can encounter such cuts before direct settlement, and whether their charged cost beats a matched, equally capable baseline. The formal results above *do not* answer those questions affirmatively.

Publication designation: PRELIMINARY / ATLAS-REPAIRED v0.2. This manuscript has undergone a fresh paper-level consistency pass after editing, but not mechanical proof verification, independent third-party refereeing, implementation, or benchmarking. It is not a certified theorem-bank receipt and does not authorize a live AMLD experiment.

References

[1] B. Tenneson, *Two Armies of One: Certified Scout–Probe Rendezvous in Proof Search*, v0.1, 26 October 2026.

[2] *ATLAS Independent Audit: Two Armies of One v0.1*, first paper-level audit, 26 October 2026.

[3] *ATLAS: Fresh Mathematical Re-Audit of Two Armies of One*, audit 0.2, 26 October 2026.

[4] B. Tenneson, *Trading, Intuition, and Return*, v0.1, preliminary companion note, 26 October 2026.

[5] B. Tenneson, *OPEN. AKM*, v0.2, preliminary theoremgraph-reversal companion, October 2026.